Causal Inference

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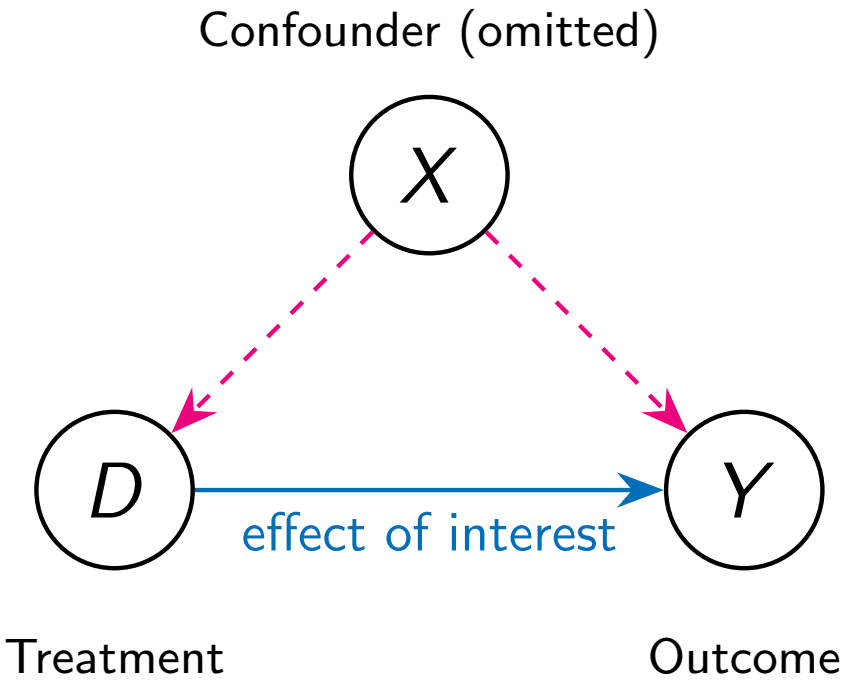
Omitted variable bias

- Regression gives us a description of the CEF, but CEF only gives us a causal relationship if all backdoor paths are closed by conditioning
- In practice, hard to know if our set of controls includes all relevant confounding variables



Reasoning about bias

- DAGs are yes-no propositions: did you condition correctly?
- In practice, answer is likely 'probably?', not 'yes'/'no'
- Going to make the meaning of 'probably' precise by asking: How sensitive is our estimate of the causal effect to an unobserved confounder?
- Ingredients for this:
  - A statistical model summarizing the relationship between treatment, potential confounders, and outcome
  - A deeper understanding of the mechanics of that model
  - Real-world knowledge about the relative strength of observed and unobserved confounders

Omitted variable bias in a regression

- The model we want is: $Y_i = \alpha + \rho D_i + \theta X_i + e_i$
- But we only have: $Y_i = \lambda + \beta D_i + u_i$
- How does leaving out X_i bias our estimated β, $\frac{Cov(Y_i, D_i)}{Var(D_i)}$?

$$\begin{aligned}
Cov(Y_i, D_i) &= Cov(\alpha + \rho D_i + \theta X_i + e_i, D_i) \\
&= Cov(\alpha, D_i) + Cov(\rho D_i, D_i) + Cov(\theta X_i, D_i) + Cov(e_i, D_i) \\
&= \rho\, Cov(D_i, D_i) + Cov(\alpha, D_i) + Cov(\theta X_i, D_i) \\
&= \rho\, Var(D_i) + \theta Cov(X_i, D_i)
\end{aligned}$$

$$\frac{Cov(Y_i, D_i)}{Var(D_i)} = \rho + \theta \times \underbrace{\frac{Cov(X_i, D_i)}{Var(D_i)}}_{\text{What is this?}} = \rho + \underbrace{\delta}_{impact} \times \underbrace{\gamma}_{imbalance}$$

Omitted variable bias (continued)

- Same holds if we include other covariates Z in the observed model and think of δ and γ as regression coefficients after residualizing wrt Z
- For imbalance:
  - Take the leftover variation of our unobserved confounder after taking out the linear relationship with Z.
  - Then ask: what is the strength of the relationship between that confounder and the treatment

$$\frac{Cov(\tilde{Y}_i, \tilde{D}_i)}{Var(\tilde{D}_i)} = \rho + \underbrace{\tilde{\delta}}_{impact} \times \underbrace{\tilde{\gamma}}_{imbalance}$$

Thinking about confounding in terms of R^2

$$R^2 = \frac{Var(\hat{Y})}{Var(Y)} = corr(Y, \hat{Y})^2$$

- R^2 measures the proportion of variance in the outcome explained by variance in the predicted outcome from the regression
- Positive characteristics: not dependent on scale of outcome
- Negative characteristics: you can always increase R^2 by putting more variables into your regression
- We can express OVB in terms of partial R^2

$$|\tilde{\delta} \times \tilde{\gamma}| = \widehat{\text{se}}(\rho)\sqrt{\frac{R^2_{Y\sim Z|D,X}\ R^2_{D\sim Z|X}}{1 - R^2_{D\sim Z|X}}}\ (\text{df}).$$

Thinking about confounding in terms of R^2

- How do we think about each term here intuitively? Which ones can we benchmark against observed confounders?

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- Some questions:
  - What increases and decreases bias?
  - What happens to $R^2_{Y\sim Z|D,X}$ as I include more control variables in my regression?

Application: Sensitivity Analysis

Example: Support for the radical right in German elections

Starting in the 2010s, vote shares for radical right-wing parties have been increasing across Europe. The change has been most pronounced in rural areas.

- Research Question:
- Does membership in a community that was historically been peripheral to the nation state cause modern individuals to vote for radical right parties?

Regression controls

Country-level:

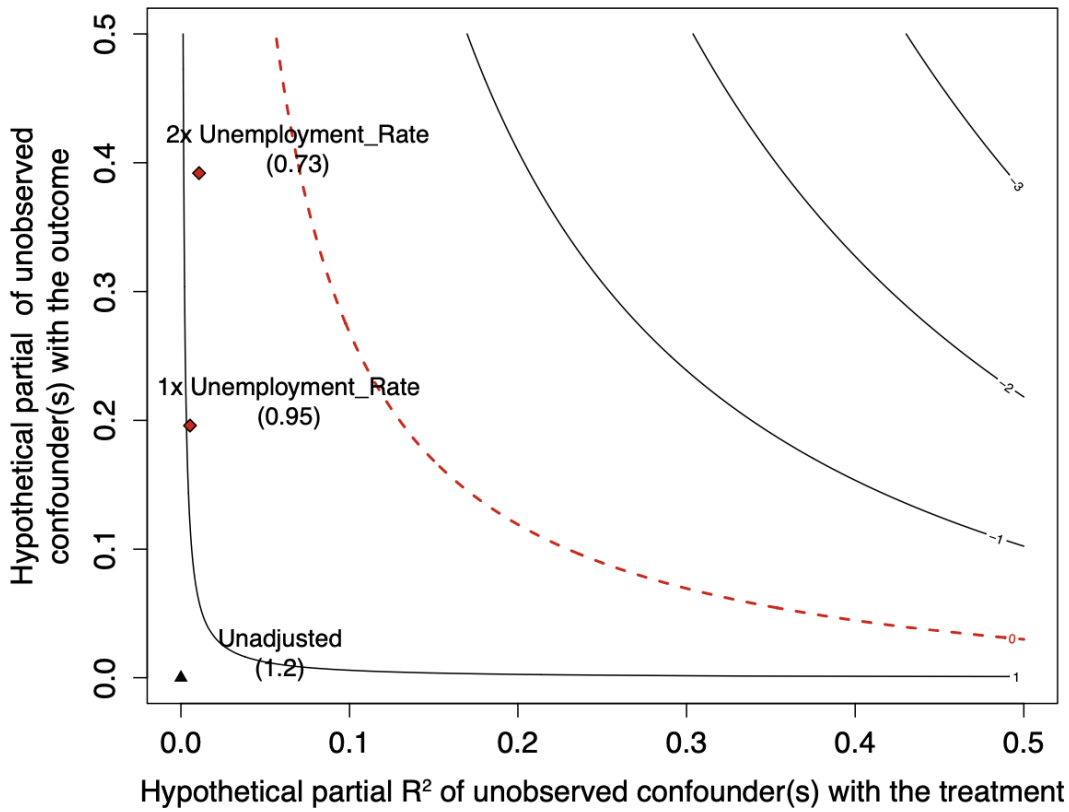
fixed effects γ_j.[8] For all county-level models, we control for regional GDP/capita, average wages, population density, unemployment rates, total population, % Catholic, the share of commuters, the physical distance to the state capital, and the CDU/CSU vote share in the 2013 general election. We provide summary statistics

Individual-level:

electoral district are stronger. We control for respondents' gender, age, education, employment status, income, the urbanity of their location, and separately also for their self-rated nationalism. The addition of the

Sensitivity analysis results

FIGURE 4. Sensitivity Analysis



Notes: Results from the sensitivity analysis proposed by Cinelli and Hazlett (2020). The plot indicates how strongly confounders would have to be correlated with the treatment and the outcome to reduce the estimated effect size to zero (dashed red line). The original effect size from Table 1 is shown in the bottom-left corner—the scenario where there is no unmeasured confounding. The red diamond shape indicates partial correlations for unemployment rates and a hypothetical confounder that is twice as strong as the unemployment rate. The interpretation of the figure is as follows: if we were to include a confounder as strong as the unemployment rate, the estimated effect size would drop to 0.95. For a confounder that is twice as strong as the unemployment rate, the estimated effect of dialectal distance on radical-right voting would drop to 0.73.